

Human Review: Zero APEPs in Infinite ℓ_1 -Sums

Original automatic proof: `solution_packet.pdf`

Checked date: 17 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proof appears to give a correct negative answer to Question 8.1 of the source paper. The construction takes

$$X = \left(\bigoplus_{n=1}^{\infty} \ell_2^n \right)_1.$$

Each coordinate space is finite-dimensional, so 0 is not an APEP of its unit ball: weak and norm topologies coincide, and a sufficiently small norm-neighbourhood of 0 in the ball cannot contain a non-empty slice.

The interesting part is the proof that 0 is nevertheless an APEP of B_X . The argument essentially leverages the finite-dimensional structure of the summands together with the fact that a weak neighbourhood of 0 in B_X is controlled by only finitely many functionals. Given such a neighbourhood, determined by m functionals, one moves to a coordinate $n > m$. In that coordinate, the m corresponding coordinate functionals have a non-trivial common kernel. Thus one can choose a unit vector $u \in \ell_2^n$ invisible to all of them.

The slice is then chosen using the support functional associated to u . Elements in this slice must have almost all of their ℓ_1 -sum norm concentrated in the n -th coordinate. Consequently the other coordinates contribute very little to the finitely many functionals defining the weak neighbourhood. In the n -th coordinate, the chosen vector u lies in the common kernel, and the Hilbert-space estimate

$$\langle h, u \rangle > 1 - \delta, \quad \|h\| \leq 1 = \|u\| \quad \implies \quad \|h - u\| \leq \sqrt{2\delta}$$

shows that the n -th coordinate is close to u . Hence the contribution from the n -th coordinate is also small. This proves that the slice lies inside the original weak neighbourhood.

Extension to Other Finite-Dimensional Blocks

The proof is not special to Hilbert spaces in its overall structure. The same argument should work with

$$X_n = \ell_p^n$$

for every fixed $1 < p < \infty$. The only replacement needed is for the Hilbert-space estimate above. In ℓ_p^n , choose

$$u \in S_{\ell_p^n} \cap \bigcap_{j=1}^m \ker a_{j,n}$$

and then choose a norming functional $u^* \in S_{(\ell_p^n)^*}$ with $u^*(u) = 1$. Since ℓ_p^n is strictly convex for $1 < p < \infty$, the slice

$$\{h \in B_{\ell_p^n} : u^*(h) > 1 - \delta\}$$

shrinks to u as $\delta \downarrow 0$. In finite dimension this can be justified by compactness. Therefore the same estimates go through after replacing the explicit Hilbert bound by this qualitative slice-shrinking fact.

For $p = 1$, the same argument does not apply, since the unit ball has flat faces and a support functional need not expose a single vector. Thus the natural extension of the proof is to the range $1 < p < \infty$.

Additional Note: Infinite-Dimensional Summands

If one wants the coordinate spaces themselves to be infinite-dimensional, the same construction should admit a straightforward modification. Take

$$X_n = \ell_2^n \oplus_1 c_0$$

and

$$X = \left(\bigoplus_{n=1}^{\infty} X_n \right)_1.$$

The ℓ_2^n -part still provides the same finite-dimensional escape direction. Given a weak neighbourhood of 0 in B_X determined by finitely many functionals, choose n larger than the number of constraints and choose

$$u \in S_{\ell_2^n}$$

orthogonal to the ℓ_2^n -coordinate parts of those functionals. Define the slice using only the ℓ_2^n -component,

$$\varphi((h_k, y_k)_k) = \langle h_n, u \rangle.$$

As before, the condition $\varphi(x) > 1 - \delta$ forces almost all of the ℓ_1 -sum mass to lie in the n -th ℓ_2^n -component. It also forces the c_0 -part inside the n -th coordinate, and all other coordinates, to have small total norm. The same Hilbert-space estimate then puts the slice inside the prescribed weak neighbourhood.

The c_0 -summand is only used to make each X_n infinite-dimensional while keeping 0 visibly non-APEP in each coordinate. One can witness this with a weak neighbourhood in B_{X_n} which controls both components, for instance

$$W_n = \{(h, y) \in B_{\ell_2^n \oplus_1 c_0} : \|h\|_2 < 1/2, |y_1| < 1/2\}.$$

No non-empty slice of B_{X_n} is contained in W_n : if the ℓ_2^n -part of the defining functional dominates, the slice meets a point of the form $(h, 0)$ with $\|h\|_2 = 1$; if the c_0 -part dominates, the slice meets a point of the form $(0, y)$ with $|y_1| = 1$, using the standard ℓ_1 -description of c_0^* . Thus 0 is not an APEP of B_{X_n} for any n , while the ℓ_1 -sum still has 0 as an APEP.

Review Outcome

Archive this packet as an apparently correct counterexample. The proof's central mechanism is sound: every coordinate fails APEP at 0 because it is finite-dimensional, while the infinite ℓ_1 -sum allows one to escape any finite list of weak constraints by moving to a sufficiently high-dimensional coordinate and choosing a direction annihilated by those constraints. The same construction should also work with ℓ_p^n blocks for $1 < p < \infty$.